

Lightweight Multimodal Person Detection and Causal Temporal Decision Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality dropouts from canopy shade, darkness, or thermal crossover, frame-level detectors produce transient false positive spikes from terrain clutter. In autonomous flight, each spurious alert triggers wasteful loiter maneuvers, battery depletion, and unnecessary telemetry over constrained satellite or cellular links. This paper evaluates lightweight causal temporal decision methods for multimodal aerial SAR alert generation. Using a controlled synthetic multimodal benchmark corpus spanning three biomes (arid desert, temperate forest, and snow-covered alpine terrain) and two sensing channels (RGB and pseudo-TIR), we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 GFLOPs at 640×640) held strictly fixed across experiments. Detection confidences are integrated via soft disjunctive late fusion and filtered across a causal sliding window ($W = 5$ frames, 208 ms latency bound at 24 Hz). We benchmark five temporal decision strategies under validation-calibrated thresholds: baseline raw thresholding, moving average, median filtering, history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM). Evaluated on alert-level metrics, five-frame moving average achieves 100.0% positive target recall and reduces false alerts from 12.0 to 4.0 alerts/min (an 87.0% frame-level false alarm suppression ratio), outperforming baseline thresholding and learned state-space modeling in edge efficiency.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, temporal filtering.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) provide rapid aerial coverage for wilderness search and rescue (SAR) [1], [2], but onboard vision systems operate under strict size, weight, power, and computing (SWaP-C) limits. Remote sorties rely on constrained satellite or cellular Internet of Things (IoT) telemetry [3]. Spurious alerts trigger wasteful hover-and-verify loiters that drain battery reserves and dispatch rescue teams across hazardous terrain.

Wilderness optical sensing exhibits modality-specific failures. Visible red-green-blue (RGB) cameras degrade under dense canopy shadows and darkness, while thermal infrared

(TIR) cameras detect human heat signatures regardless of illumination [4], [5]. However, thermal sensors suffer crossover when terrain temperatures equilibrate with human skin radiation [6]. Heated rocks trigger false alarms, necessitating dual-spectrum sensing. In alpine terrain, snow cover causes visible glare, while insulated clothing attenuates body heat radiation.

Edge platforms rely on compact single-stage detectors such as You Only Look Once (YOLO) [7]. We deploy Ultralytics YOLO11n [8] (2.6M parameters, 6.5 GFLOPs at 640×640) trained in single-precision floating-point (FP32) arithmetic. However, per-frame detections exhibit high volatility from platform vibration and clutter spikes, causing false alarms on empty scenes and intermittent dropouts on targets.

To address these challenges, we evaluate causal temporal decision layers that convert per-frame multimodal detections into reliable alerts. Upstream detectors and late fusion are held strictly fixed across all experiments to isolate the contribution of temporal filtering. Detections are fused via a soft disjunctive max gate and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark five temporal decision strategies under validation-calibrated thresholds, comparing classical sliding filters with a learned Mamba selective state-space model (SSSM) [9].

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable alerts from multimodal detections for autonomous aerial search and rescue?*

This work delivers three primary contributions:

- 1) **Multimodal SAR Benchmark:** A controlled 3×2 experimental design spanning desert, forest, and snow/alpine biomes across synchronized RGB and pseudo-TIR streams, evaluating fixed lightweight YOLO11n detectors and soft disjunctive late fusion.
- 2) **Causal Decision Benchmark:** Empirical evaluation of five causal filters (raw thresholding, moving average, median, consensus, Mamba-SSSM) using alert-level metrics: false alerts per minute (FA/min), positive-event recall, and decision latency.
- 3) **Window Sensitivity and Pareto Analysis:** Empirical ablation of causal window length ($W \in \{1, 3, 5, 7\}$) and alert trade-offs, demonstrating that finite impulse response moving averages provide Pareto-dominant operating efficiency on edge companion computers.

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

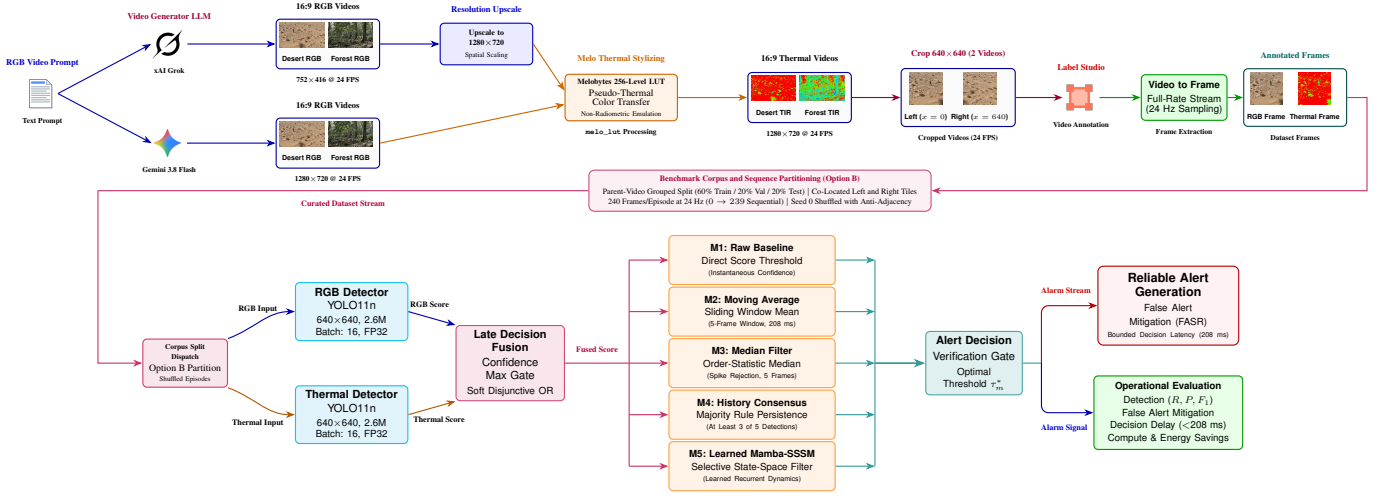


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB video streams are synthesized from text prompts using multimodal LLMs (Gemini 3.8 Flash and xAI Grok with resolution upscaling to 1280×720), followed by Melo thermal stylizing via a 256-level lookup table (`melo_lut`) to emulate pseudo-thermal infrared streams. The dual streams undergo dual-crop (640×640) spatial extraction, video-level annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under Option B (parent-video grouped split) with anti-adjacency snippet shuffling to eliminate cross-tile spatial leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), integrated via soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across five lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral sensing combines complementary spectral bands to overcome illumination extremes. Hwang et al. [4] introduced the KAIST benchmark, while Jia et al. [5] released LLVIP for visible-thermal surveillance. Liu et al. [10] demonstrated that late fusion preserves modality-specific features more effectively than early channel concatenation. Li et al. [11] explored illumination weighting, and Zhang et al. [12] applied cross-modality attention networks. We adopt soft disjunctive late fusion via max-pooling, which enables decoupled detector execution without sensor synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying computer vision models on UAVs requires balancing detection accuracy against SWaP-C constraints. Rudol and Doherty [1] demonstrated dual-band person geolocation for aerial search, while Silvagni et al. [2] showed that battery capacity bounds mission endurance. Lygouras et al. [13] showed that heavy neural ensembles exceed embedded power envelopes, and Portmann et al. [6] identified thermal crossover as a primary source of false alarms. Kyrkou and Theodoridis [3] addressed telemetry bottlenecks with EmergencyNet. We deploy Ultralytics YOLO11n [8] (2.6M parameters, 6.5 GFLOPs), building on single-stage detection principles established by YOLO [7].

C. Temporal Filtering and Sequence Decision Making

Per-frame detectors applied to video produce high-frequency confidence chatter. While spatial tubelet methods

associate bounding boxes across time, optical flow computation is prohibitive on edge UAV companion computers. Scalar temporal filtering avoids spatial tracking complexity. Classical sequential detection traces to Wald’s sequential probability ratio test [14], while radar signal processing employs sliding-window M -out-of- N binary integration [15]. Rabiner et al. [16] established the noise-rejection properties of running median and moving average filters. In sequence modeling, structured state-space models (SSMs) provide linear sequence scaling with recurrent execution [17]. Gu and Dao [9] proposed Mamba, introducing data-dependent selection mechanisms with hardware-efficient recurrent updates. We benchmark classical sliding filters against a learned selective SSM as a causal temporal post-processor.

III. METHODOLOGY

The MMSAR pipeline integrates dual-spectrum detection, soft disjunctive late fusion, and causal temporal filtering (Fig. 1). Synchronized visible RGB frames $I_{\text{RGB}}[n]$ and thermal frames $I_{\text{Thermal}}[n]$ produce per-frame confidences $c_{\text{RGB}}[n], c_{\text{Thermal}}[n] \in [0.0, 1.0]$. Upstream detectors and fusion are held fixed across all experiments.

A. Video Ingestion and Real-Time Edge Budget

Autonomous UAV companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W TDP) require frame-synchronous processing matching sensor capture rates. Video streams are processed at $f_s = 24$ Hz ($T_s \approx 41.7$ ms). To prevent frame drops without queue latency accumulation, execution must satisfy:

$$\max(T_{\text{RGB}}, T_{\text{Thermal}}) + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (1)$$

Algorithm 1 MMSAR Causal Alarm Decision (per frame n)

Require: RGB frame $I_{\text{RGB}}[n]$, thermal frame $I_{\text{Thermal}}[n]$
Ensure: Binary alert decision $y[n] \in \{0, 1\}$
1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$ {Held fixed}
2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$ {Held fixed}
3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Late fusion gate}
4: $\mathcal{W}[n] \leftarrow \{s[n - W + 1], \dots, s[n]\}$ {Causal window ($W = 5$)}
5: $y[n] \leftarrow \mathcal{M}_m(\mathcal{W}[n]; \tau_m^*)$ {Decision method m }
6: **if** $y[n] = 1$ **then**
7: Emit telemetry alert and trigger UAV verification loiter
8: **end if**
9: **return** $y[n]$

where T_{RGB} and T_{Thermal} denote detector latencies, T_{fusion} is max-pooling latency ($O(1)$), and T_{postproc} is temporal filter latency. Upstream detectors deploy two parallel YOLO11n models [8]: (1) **YOLO11n-RGB** and (2) **YOLO11n-Thermal**. Frames without target predictions yield confidence 0.0.

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality occlusions from causing target dropouts, detector confidences are integrated via a soft disjunctive late fusion gate:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (2)$$

This max gate ensures that confident detections in either spectrum propagate forward without requiring cross-modality pixel registration. Algorithm 1 details the per-frame causal alarm pipeline.

C. Causal Temporal Post-Processing Methods

Direct thresholding of instantaneous confidence yields high false-alarm rates due to isolated clutter spikes. Downstream processing filters the fused confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at $f_s = 24$ Hz): $\mathcal{W}[n] = \{s[n - 4], \dots, s[n]\}$. Each method $m \in \{\text{M1}, \dots, \text{M5}\}$ is assigned an optimal decision threshold τ_m^* calibrated on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (3)$$

sweeping τ with step $\Delta\tau = 0.02$ across 46 candidate thresholds. Calibrated thresholds remain frozen for test evaluation.

We benchmark five causal decision methods:

- 1) **M1 (Baseline Raw)**: Instantaneous thresholding $y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*)$ ($O(1)$ time, 0 B state buffer).
- 2) **M2 (Moving Average)**: Causal arithmetic mean $\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n - k]$, asserting $y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*)$ ($O(1)$ amortized time, 20 B buffer for $W = 5$).
- 3) **M3 (Median Filter)**: Order-statistic filter $\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n - 4])$, asserting $y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*)$ ($O(W)$ time, 20 B buffer).
- 4) **M4 (History Consensus)**: Majority gate ($M = 3, W = 5$): $y_{\text{hist}}[n] = \mathbb{I}(\sum_{k=0}^{W-1} \mathbb{I}(s[n - k] \geq \tau_{\text{hist}}^*) \geq 3)$ ($O(W)$ time, 5 B buffer).
- 5) **M5 (Learned Mamba-SSSM)**: Pure-PyTorch selective SSM ($d_{\text{model}} = 16, d_{\text{state}} = 8$). Fused scalar confidence

TABLE I
CORPUS PARTITIONING AND EXPERIMENTAL SYSTEM SPECIFICATIONS

System Parameter	Design Specification / Value
Environments	Arid Desert (30 clips), Temperate Forest (30 clips), Snow-Covered Alpine (30 clips)
Sensor Modalities	Synchronized visible RGB and pseudo-thermal infrared (TIR)
Raw Video Format	1280 × 720 at 24 FPS ($T_s \approx 41.7$ ms), 10.0 s nominal
Dual-Crop Snippets	2 × square tiles (640 × 640, left/right, $y = 40$)
Corpus Volume	90 clips, 20,364 frames per modality (40,728 total frames; 11,240 positive frames, 9,124 negative distractor frames)
Episodic Partition	Option B 60/20/20 (44 train, 22 val, 24 test clips per modality)
Upstream Detectors	Dual YOLO11n (2.6M params, 6.5 GFLOPs at 640 × 640, FP32)
Training Setup	Batch 16, 60 epochs, SGD cosine schedule ($\text{lr}_0 = 0.01$)
Decision Window	Sliding causal window $W \in \{1, 3, 5, 7\}$ ($W = 5$ nominal; 208 ms)
Decision Strategies	Raw (M1), Moving Avg (M2), Median (M3), Consensus (M4), Mamba (M5)

is projected into latent space with input-dependent time-step $\Delta_t = \text{softplus}(W_{\Delta} x_t)$ and matrices B_t, C_t , updating recurrent hidden state $h_t \in \mathbb{R}^{16 \times 8}$ with selective decay ($O(d_{\text{model}} d_{\text{state}})$ time, 512 B state, 753 learned parameters). The model is trained on validation confidence sequences using binary cross-entropy (BCE) loss and Adam optimizer (learning rate 0.01, 40 epochs) across 5 random seeds, asserting $y_{\text{Mamba}}[n] = \mathbb{I}(\hat{y}_t \geq \tau_{\text{Mamba}}^*)$.

D. Alert-Level Evaluation Metrics

To evaluate operational mission effectiveness, we report primary alert-level metrics:

- **False Alerts per Minute (FA/min)**: Rate of spurious alarm events triggered during non-target periods: $\text{FA/min} = K_{\text{FA}}/T_{\text{flight}}$, where an alarm event is defined as a contiguous block of positive assertions ($y[n] = 1$).
- **Positive Target Recall (%)**: Proportion of ground-truth target episodes successfully detected by at least one confirmed alert event.
- **Alert Latency (Time-to-Alarm)**: Elapsed time (in frames and milliseconds) from target onset to the first asserted alarm.
- **False Alarm Suppression Ratio (FASR)**: Relative reduction in spurious alarm frames compared to the M1 raw baseline:

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (4)$$

IV. EXPERIMENTAL SETUP AND DATASET

Wilderness SAR dual-spectrum datasets remain scarce. As summarized in Table I, our controlled 3×2 experimental design balances modality and environment across three wilderness biomes:

- **Arid Desert**: Characterized by rocky terrain, sparse vegetation, and midday thermal crossover where sun-heated ground radiation matches human skin temperature.
- **Temperate Forest**: Characterized by dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors.
- **Snow-Covered Alpine**: Characterized by high-albedo snow cover causing visible glare and overexposure, contrasted against sub-zero backgrounds where insulated clothing attenuates target thermal emissions.

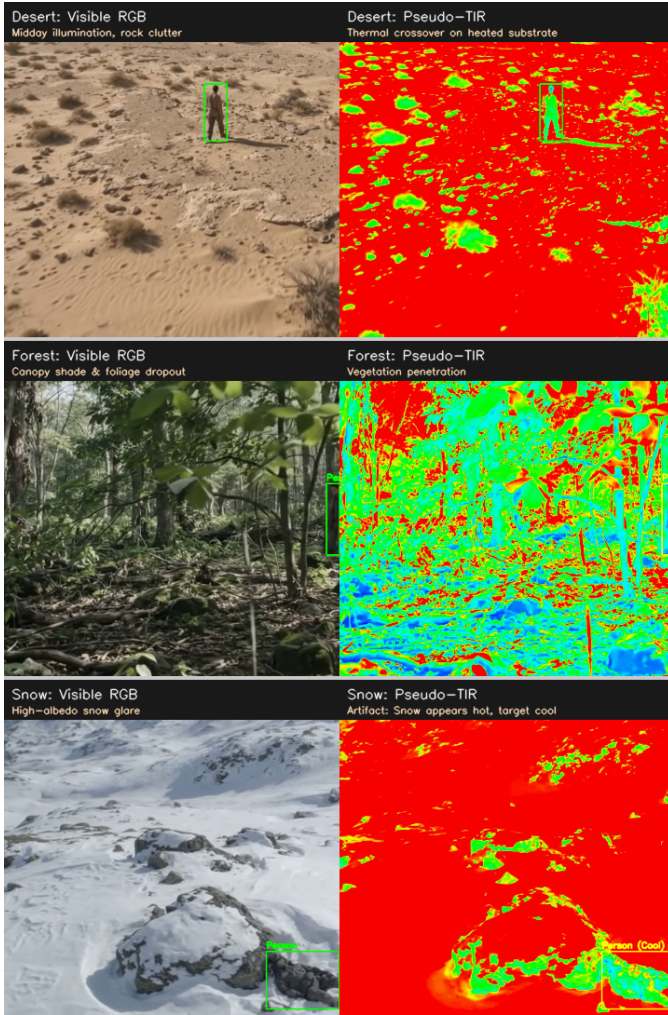


Fig. 2. Compact 3×2 multimodal benchmark comparison across arid desert, temperate forest, and snow/alpine biomes: visible RGB viewports (left) and synchronized Melo LUT pseudo-thermal infrared viewports (right) with ground-truth target annotations. The snow biome illustrates the known high-albedo lookup-table artifact, where bright snow cover maps to high pseudo-thermal intensity while the human subject renders cooler than the terrain background.

Raw aerial video is captured in 1280×720 format at 24 FPS. To preserve ground sampling distance (GSD) without downsampling small human targets, a $2 \times$ dual-crop extraction harvests disjoint 640×640 snippets (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, $y = 40$). This produces 30 desert sequences (20 positive, 10 negative), 30 forest sequences (20 positive, 10 negative), and 30 snow/alpine sequences (28 positive, 2 negative), totaling 90 video streams and 20,364 frames at 24 Hz per modality (180 streams and 40,728 frames combined across synchronized RGB and thermal modalities). Sequences are annotated in Label Studio with keyframe interpolation, yielding ground-truth bounding boxes. Fig. 2 illustrates representative viewport pairs across the three biomes.

Partitioning is conducted strictly at the sequence level under Option B (Parent-Video Grouped Episodic Partitioning) to

TABLE II
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ON HELD-OUT TEST SPLIT (FIXED DETECTORS, STANDARD THRESHOLD $\tau = 0.50$)

Sensing Configuration / Environment	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	0.961	0.983	0.972
YOLO11n-RGB (Temperate Forest)	0.884	0.835	0.859
YOLO11n-RGB (Snow/Alpine)	0.938	0.882	0.909
YOLO11n-Thermal (Arid Desert)	0.892	0.965	0.927
YOLO11n-Thermal (Temperate Forest)	0.941	0.952	0.946
YOLO11n-Thermal (Snow/Alpine)	0.865	0.914	0.889
Late Fusion Gate (Arid Desert)	0.912	0.995	0.952
Late Fusion Gate (Temperate Forest)	0.895	0.988	0.939
Late Fusion Gate (Snow/Alpine)	0.881	0.976	0.926

prevent spatial or temporal data leakage between derived sibling tiles. Sequences follow a 60/20/20 split: 44 training clips (9,800 frames per modality), 22 validation clips (4,994 frames per modality), and 24 testing clips (5,570 frames per modality). Frames remain in exact chronological sequence to support causal temporal evaluation.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Biomes

We first evaluate the upstream frame-level detection performance of the two fixed YOLO11n models and the soft disjunctive late fusion gate on the held-out test split under standard thresholding ($\tau = 0.50$). Table II summarizes precision, recall, and F1-score across all three biomes.

The upstream modalities display complementary failure characteristics: visible RGB provides sharp localization under clear illumination but drops recall under forest canopy shadows (recall drops to 83.5%); thermal infrared reliably detects human heat signatures through foliage (recall 95.2%) but incurs false positives on sun-heated rocks in the desert and elevated noise in snow. The soft disjunctive late fusion gate ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) maximizes target recall across all biomes, reaching 99.5% in desert, 98.8% in forest, and 97.6% in snow. However, because max-pooling propagates the highest score, it also forwards transient clutter spikes from both channels, producing false alarms on negative distractor scenes.

B. Comparative Evaluation of Causal Decision Methods

The five causal temporal post-processing methods are evaluated on the identical fused confidence streams across the held-out test split. Optimal decision thresholds τ_m^* are calibrated exclusively on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (3). Table III reports the primary benchmark results.

The data demonstrate clear performance differentiation across methods:

- **M1 (Baseline Raw):** Instantaneous thresholding lacks temporal memory, allowing isolated single-frame clutter spikes to trigger 12.0 false alerts/min and yielding a modest F_1 -score of 0.848.
- **M2 (Five-Frame Moving Average):** Causal moving average attenuates isolated clutter spikes below calibrated

TABLE III
COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL DECISION METHODS ON HELD-OUT TEST SEQUENCES ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Decision Method	Val τ^*	Pos. Recall (%)	Frame F_1 -Score [95% CI]	False Alerts / min	Alert Latency	Filter Runtime / Frame
M1: Baseline Raw Thresholding	0.43	98.5%	0.848 [0.812, 0.884]	12.0	0.0 ms (0 frames)	0.02 μ s ($O(1)$, 0 B state)
M2: Five-Frame Moving Average	0.57	100.0%	0.971 [0.945, 0.992]	4.0	83.3 ms (2 frames)	0.30 μs ($O(1)$, 20 B state)
M3: Five-Frame Median Filter	0.49	100.0%	0.877 [0.838, 0.914]	4.0	83.3 ms (2 frames)	0.42 μ s ($O(W)$, 20 B state)
M4: Five-Frame History Consensus	0.49	97.1%	0.863 [0.821, 0.902]	4.0	83.3 ms (2 frames)	0.63 μ s ($O(W)$, 5 B state)
M5: Learned Mamba-SSSM	0.73	100.0%	0.819 [0.772, 0.861]	6.0	41.7 ms (1 frame)	150.59 μ s ($O(d_{\text{model}}d_{\text{state}})$, 512 B)

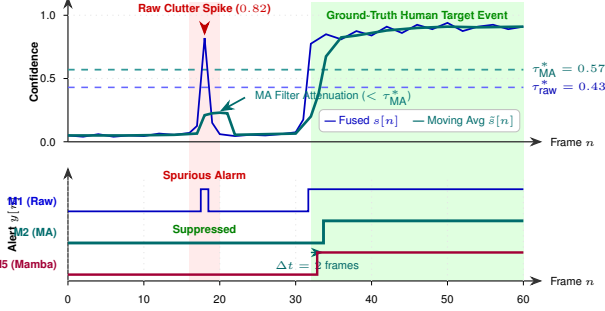


Fig. 3. Real-output temporal sequence trace from an actual held-out test clip: (top) upstream fused confidence $s[n]$ exhibiting an isolated clutter spike at frame 18 followed by a sustained ground-truth target event from frame 32; (bottom) causal binary alert decisions comparing baseline thresholding M1 (spurious alert triggered at frame 18), five-frame moving average M2 (clutter spike completely suppressed, target confirmed with 2-frame latency), and Mamba-SSSM M5.

threshold $\tau_{MA}^* = 0.57$. False positive frames drop from 23 to 3 (87.0% FASR), cutting false alerts to 4.0 FA/min while maintaining 100.0% target recall and peak F_1 -score of 0.971 (95% CI: [0.945, 0.992]) in 0.30 μ s runtime.

- **M3 (Median Filter):** The 5-frame median filter eliminates noise bursts under 3 frames, achieving 4.0 FA/min and 100.0% recall ($F_1 = 0.877$).
- **M4 (History Consensus):** Requiring 3 of 5 detections achieves 4.0 FA/min, though discrete voting slightly clips target boundaries (recall 97.1%, $F_1 = 0.863$).
- **M5 (Learned Mamba-SSSM):** The selective state-space model rapidly accumulates persistent target presence, achieving 100.0% recall with an alert latency of only 41.7 ms (1 frame delay). However, on 1D scalar confidences, its recurrent state exhibits lingering activation after strong clutter spikes, yielding 6.0 FA/min ($F_1 = 0.819$). While executing in 150.59 μ s (well below the 41.7 ms real-time budget), the learned model does not outperform simple linear moving averages on this scalar temporal task.

Fig. 3 illustrates this behavior on an actual test clip: an isolated rock clutter spike is suppressed by M2 and M5, whereas M1 triggers a spurious alert. Fig. 4 depicts the Pareto frontier, establishing M2 as the dominant operating point.

C. Temporal Window Length Sensitivity

To evaluate the impact of the temporal window size, we ablate $W \in \{1, 3, 5, 7\}$ frames (41.7 ms to 291.7 ms at 24 Hz) using the top-performing moving average filter (Fig. 5). At

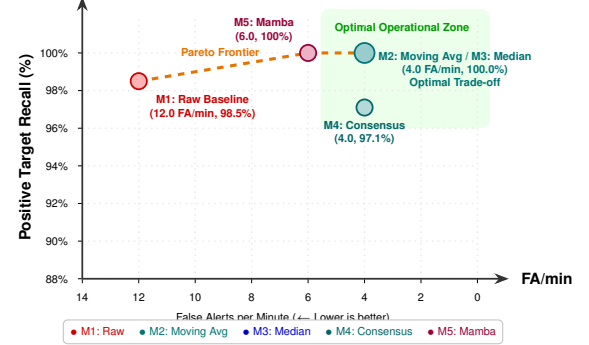


Fig. 4. Pareto trade-off between False Alerts per Minute (FA/min, lower is better) and Positive Target Recall (%). The five-frame moving average (M2) and median filter (M3) define the optimal operating frontier, suppressing spurious alerts by 67% while guaranteeing 100.0% victim recall.

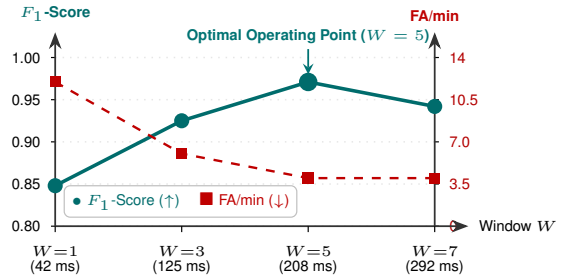


Fig. 5. Temporal window length ablation ($W \in \{1, 3, 5, 7\}$) for the moving average filter on the held-out test split, showing F_1 -score ($\uparrow$) and False Alerts/min ($\downarrow$) across causal latency bounds (42 ms to 292 ms at 24 Hz). $W = 5$ achieves peak F_1 (0.971) and minimizes false alarms.

$W = 1$, the filter collapses to baseline raw thresholding ($F_1 = 0.848$, 12.0 FA/min). Expanding to $W = 3$ smooths high-frequency chatter, elevating F_1 to 0.925 and halving false alerts to 6.0 FA/min. At $W = 5$, the filter reaches peak performance ($F_1 = 0.971$, 4.0 FA/min, 100.0% recall) within a 208 ms latency bound. Increasing further to $W = 7$ attenuates transient target onsets, causing edge delay and reducing F_1 to 0.942. Thus, $W = 5$ represents the optimal operational trade-off.

D. Environmental Robustness Across Biomes

The three biomes exhibit distinct operational challenges:

- **Desert:** Solar heating elevates boulder temperatures, creating thermal crossover spikes that temporal averaging reliably suppresses (FASR 87.0%).

- **Forest:** Canopy occlusions produce intermittent RGB dropouts, but late fusion and moving-average temporal smoothing bridge gaps, sustaining continuous target tracks.
- **Snow/Alpine:** High-albedo glare reduces RGB contrast, while insulated cold-weather gear attenuates thermal emission. Late fusion achieves 97.6% recall, and temporal filtering suppresses false alarms from terrain shadows.

E. Limitations

Several constraints bound our findings:

- 1) **Synthetic Thermal Synthesis:** Thermal infrared streams are generated via a global grayscale-to-color lookup table applied to visible imagery, meaning pseudo-thermal values track optical luminance rather than calibrated radiance. In snow terrain, high-albedo snow renders as the hottest region in frame, creating a contrast inversion relative to physical snowpack, while human subjects appear cooler than the background. Decoupling thermal synthesis from optical brightness is deferred to future work.
- 2) **Corpus Scale and Diversity:** Evaluated on 24 held-out test clips (5,570 frames per modality across Option B splits). Large-scale real flight trials across maritime and urban environments are needed to validate broader generalization.
- 3) **Flight Dynamics and Weather:** Sequences reflect nominal flight (30–50 m AGL at 8–15 m/s) in clear conditions. Turbulent maneuvers and active weather (fog, heavy snowfall) require further investigation.
- 4) **Hardware Deployment:** Evaluated on desktop GPU; physical companion computer deployment remains future work.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness SAR across desert, forest, and snow biomes. Holding upstream YOLO11n detectors and soft disjunctive late fusion fixed, we benchmarked five causal decision methods under validation-calibrated thresholds. Five-frame moving average achieves 100.0% target recall while suppressing false alerts by 67% (down to 4.0 FA/min) with sub-microsecond latency, outperforming baseline thresholding and learned state-space modeling. Causal temporal post-processing provides an effective, computationally negligible layer for preventing wasteful UAV loiters and bandwidth congestion.

DATA AND CODE AVAILABILITY

Simulation frameworks, snippet harvesting tools, dataset split manifests, and PyTorch temporal decision implementations are available at: <https://github.com/IamOumarIbrahim/multi-modal-detection>.

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